

claude-windows / 6a82308f-69d6-43a8-9ea4-d8eaf497253d

Metadata

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- Source: `claude-windows`  
- Kind: `claude`  
- Source file: `C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\subagents\workflows\wf_3d536535-241\agent-a1f01e5234c361deb.jsonl`  
- SHA-256: `46d105609306b11de9d86b1fef93cd0db8972385f79246bc8ccdc3bab92f68a0`  
- Source modified: `2026-06-16T03:21:14+00:00`  
- Imported at: `2026-07-05T16:48:27+00:00`  
- project: `wf_3d536535-241`  
- session_id: `6a82308f-69d6-43a8-9ea4-d8eaf497253d`
```

Transcript

1. user (2026-06-16T03:19:51.350Z)

Source Extractor

Research question: "Research question: **What concrete, recent (2022?2025) techniques and evaluation practices should a local, commercially-licensed badminton rally-detector adopt to close the quality gap with a strong cloud VLM (Gemini), given a tiny labeled set (~6 ? ~16 golden videos)?** Produce an adversarially-verified, cited report organized around the four dimensions below. Prefer specific named methods, papers (arXiv/venue), and reported numbers over generic advice; flag anything you cannot verify.

CONTEXT (so you target the real gap, not generic ML):

- System: a local pipeline detects the shuttle per-frame (WASB / TrackNet, MIT-licensed) ? a heuristic turns the trajectory into candidate rally windows ? today an optional Gemini "handoff" confirms rallies. We just measured a fully-local (no-AI) run vs Gemini on 3 matches: local emits far FEWER but MUCH LONGER windows (mean 65s vs 6.1s; one window spanned a whole 174s clip) ? it does NOT miss activity, it FAILS TO SEGMENT. Root cause (code-confirmed): a frame is "active" iff the shuttle is visible AND moving above a small per-second-normalized velocity threshold; active frames within a 2 s gap are merged; there is NO max-window cap and NO inactivity/boundary split. So "shuttle is moving" is conflated with "rally in progress," and dead-time + serves + knock-ups bridge rallies into mega-windows. Shuttle tracking itself is excellent (96?99% per-frame visibility).

- HARD CONSTRAINTS: weights/data/code must be MIT/Apache-2.0/BSD only (no viral/NC/custom licenses, reject Gemma-3/Llama-4-MAU-cap); we may NEVER train owned weights on paid-LLM (Gemini/OpenAI/Anthropic) outputs (terms/copyright), though paid LLMs may be used at inference; minors excluded from any training pool; runs on a single local GPU.

- WHAT WE ALREADY HAVE (do NOT re-survey the basics ? go deeper/newer): the architecture skeleton is known to be standard Temporal Action Localization/Segmentation (proposal+scorer) + late/score-level fusion + stacked generalization; in-domain prior art (MonoTrack, ShuttleSet, See et al. 2024/25 non-broadcast badminton rally start/end, Ghosh et al. point-segmentation, tennis/TT play-break state machines) is already cited. We have a working eval harness: leave-one-video-out (grouped by match), temporal-IoU F1 @0.5, F1@{0.1,0.25,0.5}, mAP@tIoU, segment-count-ratio, bootstrap 95% CIs, paired exact sign-test, Platt/isotonic calibration + Brier/ECE, a tiny numpy LogisticRegression scorer over per-window feature columns, an experiment leaderboard, and an "eval?serve" guard (a cue measured +0.074 F1 on permissive proposal windowing but ?0.008 on the coarser served windowing ? reverted). Available but default-OFF per-window signals: net-crossing count (rally gate), 5 person/court features, 3 audio-hit features.

THE FOUR DIMENSIONS:

1) WINDOWING & BOUNDARIES (Temporal Action Localization/Segmentation): beyond proposal+score ? what are the best *recent* methods for (a) turning a dense per-frame signal into well-segmented actions, (b) explicitly controlling OVER-segmentation/merging, (c) boundary detection/regression and splitting merged segments? Cover e.g. MS-TCN/MS-TCN++, ASRF (action-boundary regression), BMM/BSN boundary-matching, ActionFormer/TriDet/TemporalMaxer (actionness + boundary heads), smoothing/boundary losses, and the standard metrics for over-segmentation (segmental F1@k,

edit/Levenshtein score, mAP@tIoU). Which transfer to a 1-D shuttle-trajectory + auxiliary-cue setting on tiny data?

2) SHUTTLE-TRAJECTORY ? RALLY SIGNAL PROCESSING (racket-sports specific): how do published systems convert ball/shuttle tracking into rally/point segments and distinguish "in-play" from dead-time? Cover rally-state machines, serve/let detection, net-crossing & two-sided-occupancy cues, hit/contact (audio + kinematic) detection, and fps-invariant trajectory features. What features best separate "rally" from "shuttle merely moving"?

3) SMALL-DATA STRATEGIES: with ~6716 labeled videos, what gives the most quality per label ? active learning (which videos/segments to label next; uncertainty/diversity/core-set), semi-/weakly-/point-supervised TAL (e.g. SF-Net, timestamp supervision), self-training/pseudo-labeling, and data-efficient TAL? Separately: the legally-clean ways to exploit a strong teacher (Gemini) at INFERENCE for weak/pseudo-labels or as an evaluation oracle WITHOUT its outputs becoming training data for owned weights ? what does the literature say about teacher-student / distillation boundaries and label-noise handling, and what governance distinctions matter?

4) EVALUATION METHODOLOGY & RIGOR at small n: trustworthy measurement with ~6716 videos ? bootstrap/permutation tests, paired sign-tests, leave-one-group-out vs RL00CV bias, calibration (reliability, ECE), the over-segmentation-blindness of plain tIoU, shadow/A-B evaluation, regression gates, and concrete pitfalls (label noise, annotator agreement/IAA, multiple-comparison/over-fitting traps, eval-vs-serve drift). What metrics + protocols should a small-data sports-segmentation leaderboard standardize on?

Deliverable: a structured, cited report with, per dimension, the 375 highest-leverage techniques that APPLY to our constrained setting (small data, local, license-clean, 1-D trajectory + cues + learned scorer), each with: what it is, why it fits our over-merge/small-data problem, the expected difficulty/payoff, and the citation. Call out which ideas are low-effort high-confidence wins vs research bets. Be explicit about anything unverified or license-incompatible."

Fetch and extract key claims from this source:

****URL:**** <https://arxiv.org/pdf/2210.10352>

****Title:**** Temporal Action Segmentation: An Analysis of Modern Techniques (Ding et al., survey 2023, TPAMI)

****Found via:**** TAL boundary / over-segmentation methods (academic, recent) search

Task

1. Use WebFetch to retrieve the page content.
2. Assess source quality: primary research/institution? secondary reporting? blog/opinion? forum? unreliable?
3. Extract 2-5 FALSIFIABLE claims that bear on the research question. Each claim must:
 - be a concrete, checkable statement (not vague generalities)
 - include a direct quote from the source as support
 - be rated central/supporting/tangential to the research question
4. Note publish date if available.

If the fetch fails or the page is irrelevant/paywalled, return claims: [] and sourceQuality: "unreliable".

Structured output only.

2. assistant (2026-06-16T03:19:53.339Z)

I'll fetch the source and extract the relevant claims.

3. user (2026-06-16T03:20:04.873Z)

I cannot extract the requested claims about temporal action segmentation techniques from the provided content.

The document appears to be a PDF file containing binary/compressed data and image streams. While the file structure shows it contains some content (PDF objects, images, filters), the actual text content related to temporal action segmentation, over-segmentation problems, MS-TCN

smoothing loss, ASRF action boundary regression, or segmental F1 metrics is not readable in the provided material.

To properly analyze claims about:

- Over-segmentation control methods
- TAS evaluation metrics (segmental F1@k, edit score, frame accuracy)
- Weakly-supervised approaches
- Comparative performance numbers

I would need the document in a format where the text layer is extractable (such as a properly OCR'd PDF, plain text, or HTML version). The current representation is primarily structured as encoded streams and image data that cannot be reliably decoded without the appropriate decompression context.

****Recommendation:**** Please provide the source material as plain text, markdown, or a text-extractable PDF format.

[Binary content (application/pdf, 1.8MB) also saved to C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-178157999760-41alwg.pdf]

4. assistant (2026-06-16T03:20:07.352Z)

The PDF binary wasn't extractable via WebFetch, but it was saved locally. Let me try the abstract page on arXiv (HTML) and also read the saved PDF directly.

5. user (2026-06-16T03:20:10.194Z)

pdftoppm failed: Command 'pdftoppm' not found or is in an unsafe location (current directory)

6. user (2026-06-16T03:20:19.485Z)

Temporal Action Segmentation: An Analysis of Modern Techniques

****Title:**** "Temporal Action Segmentation: An Analysis of Modern Techniques"

****Authors:**** Guodong Ding, Fadime Sener, Angela Yao

****Publication:**** TPAMI 2023 (submitted October 19, 2022; last revised October 21, 2023)

****Abstract:**** The paper describes TAS as aiming to "densely identify video frames in minutes-long videos with multiple action classes." It provides a systematic survey examining "frame representation and temporal modeling" and reviews "existing TAS works categorized by their levels of supervision."

****Specific Claims Found:**** The abstract mentions curating "a list of TAS resources" but does not contain direct quotes about over-segmentation control, specific evaluation metrics (segmental F1@k, edit score, frame accuracy), or boundary detection methods (MS-TCN, ASRF). The document indicates the paper discusses "common benchmarks" and "prevalent evaluation measures," but these specific technical details are not quoted in the provided webpage content.

7. assistant (2026-06-16T03:20:24.903Z)

The PDF needs OCR/text extraction. Let me try to extract text from the saved PDF using Python (pdfminer or PyPDF2), which is often available.

8. user (2026-06-16T03:20:26.443Z)

pypdf ok

9. user (2026-06-16T03:20:34.981Z)

==== KW: over-segment ====

'ture. Furthermore, MoF, as a per-frame calculation, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation. \nSegment-Based Measures. The Edit Score [65] quantifies the similarity

'dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation. \nSegment-Based Measures. The Edit Score [65] quantifies the similarity of two sequences. It is based on the Levenshtein or edit distance and tallies the minimum number of insertions, deletions, and replacement op'

'hot labels adopt a step function with hard action labels assignments. (b) Linear [81] linearly mixes the action probabilities. (c) ABS [55] uses a sigmoid function with a decay proportional to the action duration. \n3.2.3 Over-Segmentation \nLocal continuity is an inherent attribute of procedural actions, meaning an action should be locally continuous and only change at its actual boundary. This has motivated researchers to refine the resulting seg'

'change at its actual boundary. This has motivated researchers to refine the resulting segments at the boundaries to improve performance. \nBoundary Refinement. Wang et al. [96], motivated to reduce boundary ambiguity and over-segmentation, proposed a refinement module for MS-TCN [14]. Their module uses a novel Local Barrier Pooling operation to smooth boundary predictions with confident ones in the later stages of MS-TCN. Separately, [97] proposed'

'each stage, the (full) temporal resolution is preserved by progressively dilated convolutions. MS-TCN significantly improves segmentation performance compared to earlier methods [52], [82] by a large margin and reduces over-segmentation errors. The follow-up MS-TCN++ [85] introduces a dual dilated layer and incorporated parameter sharing in the refinement stages. \nSeveral follow-up works proposed improvements to the MS-TCN architecture. RPGAussian '

'ASS FILES, VOL. 14, NO. 8, AUGUST 2015 \nincorporates a multi-stage sequence-level refinement loss to rectify errors from previous predictions. \nC2F-TCN [83] is an encoder-decoder framework with the motivation to tackle over-segmentation. This model incorporates a coarse-to-fine ensemble of decoding layers, resulting in less fragmented segments. This work also introduces a multi-resolution feature-level augmentation strategy and a complex activ'

'ing into an autoregressive framework to predict transcripts. Removing the frame-wise prediction allowed UVAST to outperform previous methods on the segment metrics of F1 and Edit Score. \nTo simultaneously address the over-segmentation of TCNs and Transformers requiring substantial amounts of training data, TCTr [108] proposes a hybrid approach. It consists of one convolution stage and four Transformer encoding stages to balance complexity and pe'

'uses boundary locations. \nThis approach is model-independent and applicable to many temporal segmentation output. DPRN [114] proposes a divide-and-conquer approach to maximize frame-wise classification accuracy and reduce over-segmentation errors. UARL [115] estimates the uncertainty stemming from ambiguous boundaries by Monte-Carlo sampling, while \nSEDT [116] introduces a strategy to smooth annotations around boundaries into soft labels. \n4.2 Weakly'

==== KW: oversegment ====

'hile the majority of existing techniques follow frame-wise prediction (Eq. (2)). \nWe recommend a greater emphasis on solving the task at the segment level and anticipate that it can greatly reduce the current issues with oversegmentation. \nForms of Supervision. Procedural video sequences feature enormous temporal redundancy in the supervisory signals due to the significant similarity between successive video frames of the same motion. Such redundancy

==== KW: segmental F1 ====

==== KW: edit ====

'ng [49]. However, their applicability may be limited for real-time tasks such as action anticipation or early detection. Specifically, online videos are often produced content; there may be domain gaps that arise from editing processes like fast-forwarding, annotated frames, or changing viewpoints. \nYet, the diversity of existing recorded procedural activity datasets is rather limited. With only a few small-scale exceptions [36], [37],'

fluence. The scorer ranges between 0 and 1, where 0 signifies no repetitions, and higher scores reflect a higher degree of repetition within the sequence. The order variation score, v , is defined as the normalized average edit distance, $e(X, Y)$, between a pair of sequences, (X, Y) . This score is further normalized with respect to the maximum sequence length. $v = 1 - e(X, Y) / \max(|X|, |Y|)$, (4) The score v also has a range of $[0, 1]$, where 1 indicates

indicating the most frequent action class has only 6 times the number of frames compared to the least frequent class. On the JOURNAL OF LATEX CLASS FILES, VOL. 14, NO. 8, AUGUST 2015

5 \nGT: \nPred: \nDiff: \n(a) MoF \nGT: \nPred: \nEdit \nDist.: \n(b) Edit Score \nGT: \nPred: \nDet.: \n(c) F1 Score \nFig. 5: Evaluation metric visualization. (a) MoF estimates how accurate frame wise predictions are. (b) Edit score tolerates small boundary shifts, as long as the sequence

most frequent action class has only 6 times the number of frames compared to the least frequent class. On the JOURNAL OF LATEX CLASS FILES, VOL. 14, NO. 8, AUGUST 2015

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14, NO. 8, AUGUST 2015 5 \nGT: \nPred: \nDiff: \n(a) MoF \nGT: \nPred: \nEdit \nDist.: \n(b) Edit Score \nGT: \nPred: \nDet.: \n(c) F1 Score \nFig. 5: Evaluation metric visualization. (a) MoF estimates how accurate frame wise predictions are. (b) Edit score tolerates small boundary shifts, as long as the sequence order is correct. (c) F1 score is the harmonic mean of the precision and recall of the detected action segments. The formulas can be found in Section 2.4.

eos at a time, while activity label supervision works with videos from all activities simultaneously.

2.4 Evaluation

2.4.1 Evaluation Measures

Three commonly adopted evaluation metrics in TAS are Mean over Frames (MoF), Edit Score, and F1 scores. The first is a frame-based measure, while the latter two are segment-based measures. All three metrics are used in full, weak, and semi-supervised settings. For unsupervised settings, only F1 and

on, does not capture segment quality; the score can be high even when the segments are fragmented. Dividing an action into many discontinuous sub-segments is referred to as over-segmentation. Segment-based measures like Edit-score [65] and F1-score [52] are instead more suitable measures for evaluating phenomenon like over-segmentation.

Segment-Based Measures.

The Edit Score [65] quantifies the similarity of two sequences. It is based on

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Segment-Based Measures.

The Edit Score [65] quantifies the similarity of two sequences. It is based on the Levenshtein or edit distance and tallies the minimum number of insertions, deletions, and replacement operations required to convert one segment sequence into another. Consider X and Y as the ordered list of predicted and ground truth actions

$j \geq 1 + 1 - \mathbb{I}(x_i \neq y_j)$, otherwise. (6) where $i \in [1, |X|]$, $j \in [1, |Y|]$ are indices for X and Y , respectively, and $\mathbb{I}(\cdot)$ is the indicator function. Eq. (6) can be effectively solved by dynamic programming. The final edit distance value is then normalized by the maximum length of the two sequences to compute the Edit score: $\text{Edit Score} = \frac{1}{\max(|X|, |Y|)} \times \text{edit distance}$. (7) As a metric, the Edit score can assess how well a model

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$\frac{1}{|X|+|Y|} \sum_{i=1}^{|X|+|Y|} \min(|X|, |Y|)$ as a metric, the Edit score can assess how well a model predicts the sequence of actions without requiring exact frame-wise correspondence to the ground truth. The F1 score or F1@? [52] compares the Intersection over Union (IoU) of each segm'

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N lists methods built on Temporal Convolutional Networks. TF uses transformer as the backbone. Refine group aims to improve the performance of existing backbones. Method Year Input/Feature GTEA Breakfast F1@{10, 25, 50} Edit MoF F1@{10, 25, 50} Edit MoF Repr. [98] Bi-LSTM 2016 RGB + flow 66.5 59.0 43.6 - 55.5 - - - - [65] ST-CNN 2016 RGB + motion* 58.7 54.4 41.9 - 60.6 - - - - [99] LCDC 2019 RGB + motion* 52.4 - - 45.4 55.3 - - - - [1]

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ar\encoder structure but changes the decoding into an auto-regressive framework to predict transcripts. Removing the frame-wise prediction allowed UFAST to outperform previous methods on the segment metrics of F1 and Edit Score. To simultaneously address the over-segmentation of TCNs and Transformers requiring substantial amounts of training data, TCTR [108] proposes a hybrid approach. It consists of none convolution stage and four Transf'

d to boost the performance. Complex activity (CA) denotes the usage of video-level labels for training, which is only applicable to Breakfast. Method Year FL TD FE Backbone Breakfast 50Salads GTEA CA F1@{10, 25, 50} Edit Acc F1@{10, 25, 50} Edit Acc F1@{10, 25,

10. assistant (2026-06-16T03:20:38.037Z)

I have rich extractable text. Let me pull a few more passages on the smoothing loss (T-MSE), timestamp/weak supervision, and the boundary-regression refinement methods to nail down precise quotes.

11. user (2026-06-16T03:20:46.680Z)

==== KW: smoothing loss ====
 (none)
 ==== KW: truncated mean squared ====
 (none)
 ==== KW: T-MSE ====
 (none)
 ==== KW: single timestamp ====
 'Xiv preprint arXiv:1910.09217 , 2019. [52] C. Lea, M. D. Flynn, R. Vidal, A. Reiter, and G. D. Hager, 'Temporal convolutional networks for action segmentation and detection,' in CVPR, 2017. [53] D. Moltisanti, S. Fidler, and D. Damen, 'Action recognition from single timestamp supervision in untrimmed videos,' in CVPR, 2019. [54] D. Singhania, R. Rahaman, and A. Yao, 'Iterative contrast-classify for semi-supervised temporal action segmentation,' in AAAI, 2022. [55] G. Ding and A. Yao, 'Leveraging action affinity and continuity for '

KW: timestamp supervision ====
 'S FILES, VOL. 14, NO. 8, AUGUST 2015 13 TABLE 6: Performance of weakly supervised methods on Breakfast and 50Salads. Tr indicates action transcripts, T is for iterative two-stage solutions, while S is for single-stage. Action sets is denoted by Set. TS is for timestamp supervision. Method Year Feature Breakfast 50Salads MoF IoU IoD MoF Tr + T [128] HTK 2017 IDT + FV 25.9 - - 24.7 [11] HMM/RNN 2017 IDT + FV 33.3 - - 45.5 [81] ISBA 2018 IDT + FV 38.4 24.2 40.6 - [42] TASL 2021 IDT + FV 49.9 36.6 34.3 - Tr + S [130] ECTC 2016 IDT + FV 27.'

videos, action pairs intend to demonstrate a consistent temporal order. To leverage this insight, POC introduces a pairwise order consistency loss that penalizes disagreements in ordering between the extracted templates and the segmentation model outputs. 4.2.3 Timestamp

Supervision\nTimestamp supervision provides action labels for a sparse\nset of frames instead of the dense video sequence. Some\nmethods [136], [137] place constraints on this set, i.e. one\nframe from each action (equivalent to an augmented tran-\nscript in which each action'

'end to demonstrate a consistent temporal order. To leverage\nthis insight, POC introduces a pairwise order consistency\nloss that penalizes disagreements in ordering between the\nextracted templates and the segmentation model outputs.\n4.2.3 Timestamp Supervision\nTimestamp supervision provides action labels for a sparse\nset of frames instead of the dense video sequence. Some\nmethods [136], [137] place constraints on this set, i.e. one\nframe from each action (equivalent to an augmented tran-\nscript in which each action is matched to a singl'

'es either leverage the unlabeled set for better rep-\nresentation or mine priors from them. Detailed performance\ncomparisons are given in Tab. 8.\nSemiTAS [55] demonstrates that even a small subset\nof densely annotated videos provides more informative\ncues than timestamp supervision applied to the entire\ndataset. SemiTAS shows that such supervision offers valu-\nable action-level priors to guide learning on unlabeled\nvideos. It introduces two novel loss functions for semi-\nsupervised TAS: the action affinity loss and the action\ncontinuity'

'print arXiv:1910.09217 , 2019.\n[52] C. Lea, M. D. Flynn, R. Vidal, A. Reiter, and G. D. Hager,\nTemporal convolutional networks for action segmentation and\ndetection, in CVPR, 2017.\n[53] D. Moltisanti, S. Fidler, and D. Damen, Action recognition from\nsingle timestamp supervision in untrimmed videos, in CVPR, 2019.\n[54] D. Singhania, R. Rahaman, and A. Yao, Iterative contrast-classify\nfor semi-supervised temporal action segmentation, in AAAI, 2022.\n[55] G. Ding and A. Yao, Leveraging action affinity and continuity\nfor semi-supervi'

'd temporal action segmentation, in ECCV, 2022.\n[56] J. Li and S. Todorovic, Action shuffle alternating learning for\nunsupervised action segmentation, in CVPR, 2021.\n[57] R. Rahaman, D. Singhania, A. Thiery, and A. Yao, A generalized\n& robust framework for timestamp supervision in temporal\naction segmentation, in ECCV, 2022.\n[58] J. Li, P. Lei, and S. Todorovic, Weakly supervised energy-based\nlearning for action segmentation, in ICCV, 2019.\n[59] M. Fayyaz and J. Gall, Sct: Set constrained temporal transformer\nfor set supervised'

'iscriminative proto-\ntypes with dynamic time warping, in CVPR, 2021.\n[135] J. Li and S. Todorovic, Anchor-constrained viterbi for set-\nsupervised action segmentation, in CVPR, 2021.\n[136] Z. Li, Y. Abu Farha, and J. Gall, Temporal action segmentation\nfrom timestamp supervision, in CVPR, 2021.\n[137] H. Khan, S. Hareesh, A. Ahmed, S. Siddiqui, A. Konin, M. Z.\nZia, and Q.-H. Tran, Timestamp-supervised action segmentation\nwith graph convolutional networks, in IROS, 2022.\n[138] Y. Sourì, Y. A. Farha, E. Bahrami, G. Francesca, and J. G'

'resh, A. Ahmed, S. Siddiqui, A. Konin, M. Z.\nZia, and Q.-H. Tran, Timestamp-supervised action segmentation\nwith graph convolutional networks, in IROS, 2022.\n[138] Y. Sourì, Y. A. Farha, E. Bahrami, G. Francesca, and J. Gall,\nRobust action segmentation from timestamp supervision, in\nBMVC, 2022.\n[139] M. A. Fligner and J. S. Verducci, Distance based ranking mod-\nels, Journal of the Royal Statistical Society. Series B (Methodological) ,\npp. 359-369, 1986.\n[140] K. Goel and E. Brunskill, Learning procedural abstractions and\nevaluati'

==== KW: boundary regression ====

(none)

==== KW: ASRF ====

'8 80.0 74.2 68.6 56.5 73.6 71.0\n[111] SSTDA + MS-TCN 2020 I3D 90.0 89.1 78.0 86.2 79.8 75.0 69.1 55.2 73.7 70.2\n[79] GTRM + MS-TCN** 2020 I3D - - - - 57.5 54.0 43.3 58.7 65.0\n[96] BCN + MS-TCN 2020 I3D 88.5 87.1 77.3 84.4 79.8 68.7 65.5 55.0 66.2 70.4\n[97] ASRF + MS-TCN 2020 I3D 89.4 87.8 79.8 83.7 77.3 74.3 68.9 56.1 72.4 67.6\n[112] G2L + MS-TCN 2021 I3D 89.9 87.3 75.8 84.6 78.5 74.9 69.0 55.2 73.3 70.7\n[95] FIFA + MS-TCN 2021 I3D - - - - 75.5 70.2 54.8 78.5 68.6\n[95] FIFA + UVAST [91] 2022 I3D 82.9 79.4 64.7 90.'

'ance the segmentation results.\nSeveral post-processing approaches target improving the\naction boundaries. BCN [96] introduces a module for MS-\nTCN that features a pooling operator to smooth noisy\nlow-confidence boundary predictions with clean, confident\nones. ASRF [97] refines boundaries with a complementary\nnetwork branch that directly regresses boundary locations.\nThis approach is model-independent and applicable to\nmany temporal segmentation output. DPRN [114] proposes\na divide-and-conquer approach to maximize frame-'

==== KW: Action Segment Refinement ====

(none)

==== KW: generate ====

'onstrained optimization by linear approxi-\nmation (COBYLA) [93]. The explicit modeling of lengths\nis necessary to avoid producing unreasonably long action\nsegments.\n\nThe third component, the visual model , provides the\nprobability of a feature sequence x being generated by the\ngiven action labels c . There are multiple ways to model the\nframe likelihood. For example, [12] follows Bayes? theorem\nand estimates $p(xt|cn)$ by considering:\np($xt|cn$) ? $p(cn|xt)\np(cn)$. (16)\n\nThe prior $p(cn)$ can be estimated either empirically, based\no'

'tify effective combinations of receptive\nfields for existing segmentation models. A recent work, Dif-\nfAct [117] using an ASFormer [90] backbone, presents state-\nof-the-art performance using denoising diffusion models.\n\nThe segmentation outputs are iteratively generated from\nrandom noise conditioning on the input video features. Ad-\n\nAdditionally, DiffAct proposes a masking strategy that jointly\nuses position, boundary, and relation priors of human ac-\ntions to enhance the segmentation results.\n\nSeveral post-processing approach'

'. ECTC [130] extends the temporal classi-\nfication from [131] to align transcripts with video frames\nwhile imposing consistency restrictions. It enforces frame-\nwise similarities to ensure consistent action alignments.\n\nNN-Viterbi [12] uses Viterbi decoding to generate\npseudo-labels from transcripts to train their framework\ncomposed of visual, context and length models. It presents a\nsignificant performance advancement compared to previous\nmethods. D 3TW [132] uses a differentiable alignment loss\n\nJOURNAL OF LATEX CLASS FILE'

'75.9\n[137] GCN-TSS 2022 I3D 61.4 - - 75.1\n[138] RAS-TSS 2022 I3D 62.9 - - 79.3\n\ninto model positive and negative transcripts discriminatively.\n\nSimilarly, CDFL [58], which builds upon NN-Viterbi, uses\nindiscriminative transcript modeling. However, unlike D3TW,\nCDFL generates valid and invalid candidates using a seg-\nmentation graph; invalid candidates violate the transcripts.\nCDFL also introduces a new loss based on energy differ-\nences between the valid and invalid candidates using a re-\ncursive estimation of each candidate's s'

' action sets. Building upon [125], their\nframework leverages a context, length and action model\ninto find the sequence of actions that maximizes the overall\nlikelihood for a given video, which can be solved via Viterbi.\n\nTo limit the search space, this framework generates candi-\ndate transcripts using a context-free grammar; the problem\nis effectively transformed into the action transcript setting\nwith multiple transcripts. However, it should be noted that\ncandidate transcripts may not fully cover all possible action\nsequence'

'ccuracy. SCT [59] learns\na segmentation network directly from the action sets using\na set prediction loss. Their model initially segments videos\ninto regions, followed by action probabilities and tempo-\nral lengths estimation using one branch. A second branch\ngenerates frame-wise action predictions; the two branches\nare linked with a consistency loss between frame-wise and\nregion predictions.\n\nSCV [60] uses a set-constrained Viterbi to generate\npseudo ground truths and an n-pair loss to minimize the\ncosine distance between '

'robabilities and tempo-\nral lengths estimation using one branch. A second branch\ngenerates frame-wise action predictions; the two branches\nare linked with a consistency loss between frame-wise and\nregion predictions.\n\nSCV [60] uses a set-constrained Viterbi to generate\npseudo ground truths and an n-pair loss to minimize the\ncosine distance between pairs of training videos sharing\naction classes in their respective action sets. To ensure\nthat all actions in the set are considered in the frame-wise\npseudo-ground truth, SCV ad'

'nts on this set, i.e. one\nframe from each action (equivalent to an augmented tran-\nscript in which each action is matched to a single frame),\nwhile others [57] are more relaxed and allow for arbitrarily\nsampled frames. The general strategy of timestamps is to\ngenerate pseudo frame-wise labels for supervision, then re-\nfine them iteratively. Notably, timestamp methods perform\ncomparably with fully supervised approaches, making it a\ncompelling direction for further exploration.\n\nLi et al . [136] introduced a novel approach to'

' aligned to the\nvisual context for all videos. Unfortunately, this assumption\ndoes not always hold and could be completely void.\n\nSener et al . [153] proposed a hybrid generative model\nthat combines visual and language cues for video se

12. assistant (2026-06-16T03:20:51.864Z)

I have all the precise quotes I need. I'll extract the key falsifiable claims.

This is the authoritative TPAMI 2023 survey (Ding, Sener, Yao) ? a primary academic source. The most central claims for the research question concern: (1) why frame-wise accuracy (MoF) is blind to over-segmentation while Edit/F1@? are not; (2) the named over-segmentation control methods (ASRF boundary regression, BCN, C2F-TCN); (3) timestamp supervision performing comparably to full supervision; and (4) SemiTAS's finding that a small densely-annotated subset beats timestamp supervision over the whole set.

13. user (2026-06-16T03:21:03.479Z)

Structured output provided successfully

14. assistant (2026-06-16T03:21:14.339Z)

Extracted 5 falsifiable claims from the Ding et al. TPAMI 2023 survey (primary academic source, full text recovered by parsing the locally-saved PDF with pypdf after WebFetch's markdown conversion failed on the binary).

Most relevant findings for the over-merge/small-data research question:

- The over-segmentation-blindness of frame accuracy (MoF) vs. Edit/F1@tau is stated explicitly ? directly validates the team's existing segment-count-ratio + temporal-IoU F1 leaderboard choices and argues for adding Edit/Levenshtein score.
- ASRF's model-independent boundary-regression branch is the most transferable splitting technique ? it can post-process the existing dense per-frame shuttle signal without retraining the detector.
- Two small-data labeling-strategy signals: timestamp supervision reaching near-full-supervision parity, and SemiTAS's counter-finding that a small densely-labeled subset beats timestamps spread thin ? relevant to allocating the ~6?16 golden-video budget.

Source PDF saved at: C:\Users\avidu\.claude\projects\C--Users-avidu-Projects-badminton-highlight-indexer\6a82308f-69d6-43a8-9ea4-d8eaf497253d\tool-results\webfetch-1781579999760-41alwg.pdf